

Reinforcement Learning - Sutton and Barto

Book Club Highlights

Alexandra Ortan

September 20, 2026

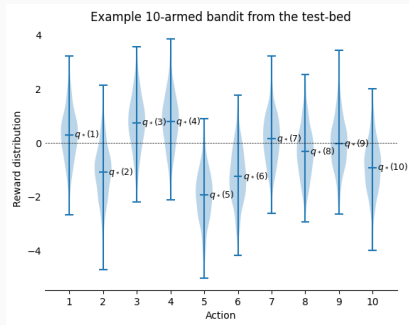
Chapter 2

Multi-armed Bandits

The 10-armed Testbed

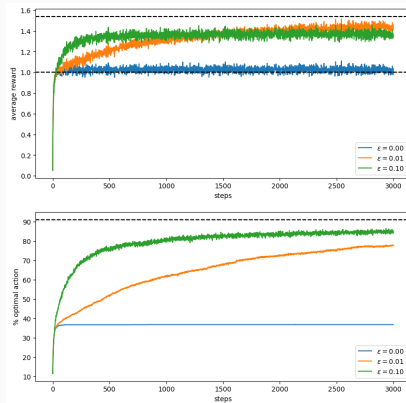
Experimental setup

- 2000 randomly generated 10-arm bandits
- $q_*(a) \sim \mathcal{N}(0, 1)$
- $R(a) \sim (q_*(a), 1)$



Experiment

- ϵ -greedy agent
- sample average value estimator
- reward at each step averaged over 2000 bandits

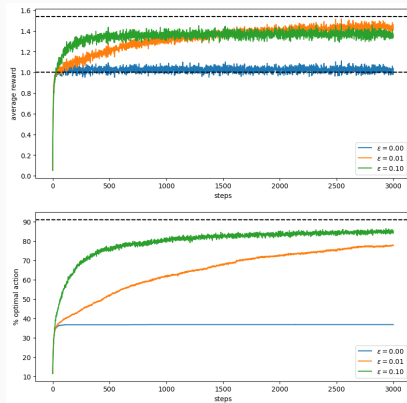


Observation 1:

For $\epsilon = 0$, average reward ≈ 1

What gives?

- This is the expected value of the first positive reward received.
- Can be estimated via integrals weighted by survival functions - the chance that a random walk stays positive forever.
- Depends on initial action value estimates (0 here).
- Value is linear in the variance of the expected reward over actions.

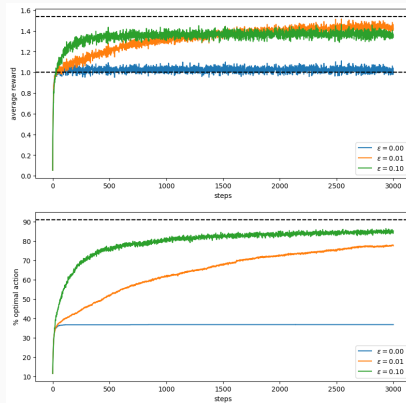


Observation 2:

Average reward $\lesssim 1.54$

What gives?

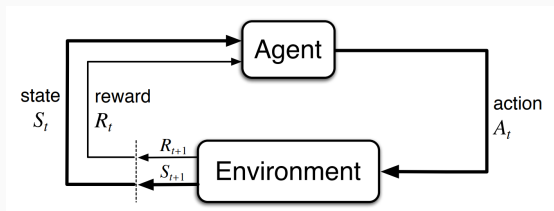
- Theoretical ceiling, as provided by an oracle agent.
- This is the best **any** agent can do.
- Calculated as the expected value of the highest of k action values.
- Ceiling $\approx \sqrt{2 \ln k}$
- For $k = 10$, this ceiling ≈ 1.539 .



Chapter 3

Finite Markov Decision Processes

Agent-Environment Interface



agent learner and decision maker

environment everything outside the agent

$S_t \in \mathcal{S}$ state of the environment at time t

$A_t \in \mathcal{A}(s)$ action at time t

$R_t \in \mathcal{R} \subset \mathbb{R}$ reward received as a consequence of action A_{t-1}

Markov Decision Processes (MDP)

Markov Property

The probability of each possible value for S_t and R_t is completely determined by the preceding state and action S_{t-1} and A_{t-1} .

Dynamics Function

The function $p : \mathcal{S} \times \mathcal{R} \times \mathcal{S} \times \mathcal{A} \mapsto [0, 1]$ is defined as

$$p(s', r \mid s, a) \doteq \Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\}$$

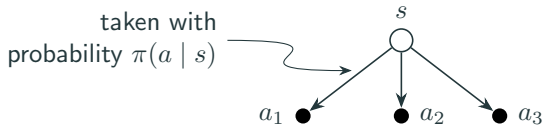
and completely defines the dynamics of the MDP.

Policies and Value Functions

Policy

Mapping from states to probability of selecting each possible action.

$$\pi : \mathcal{S} \mapsto \{\text{probabilities over } \mathcal{A}(s)\}$$

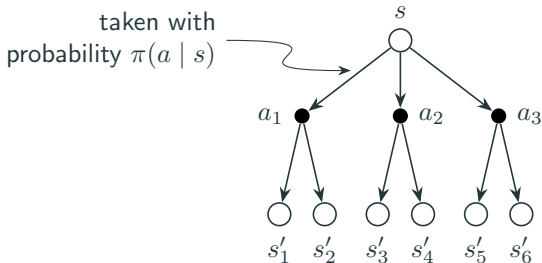


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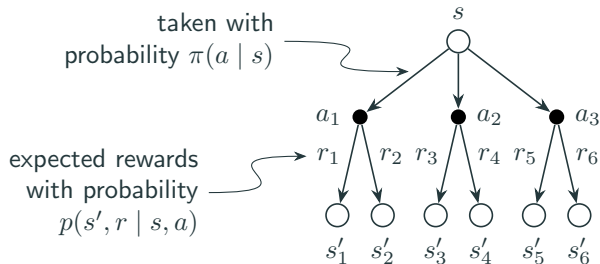


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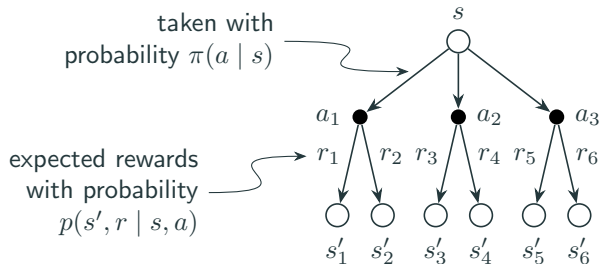


Policies and Value Functions

State value function

Expected return when starting in a given state s and following policy π thereafter.

$$v_{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid S_t = s]$$

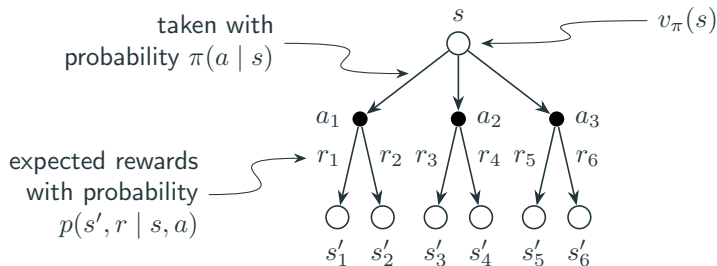


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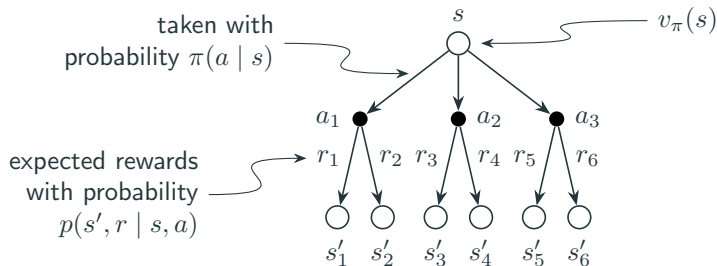


Policies and Value Functions

Action value function

Expected return when starting in a given state s , taking an action a , then following policy π thereafter.

$$q_{\pi}(s, a) = \mathbb{E}_{\pi}[G_t \mid S_t = s, A_t = a]$$

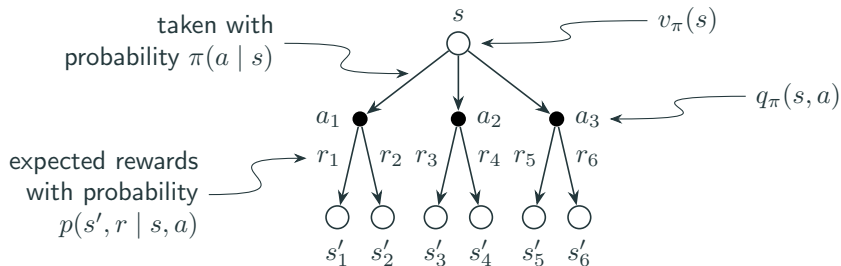


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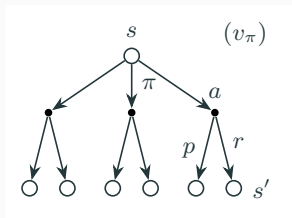
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Policies and Value Functions

Bellman Equations

$$v_{\pi}(s) = \sum_{a \in \mathcal{A}(s)} \pi(a | s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r | s, a) [r + \gamma v_{\pi}(s')]$$

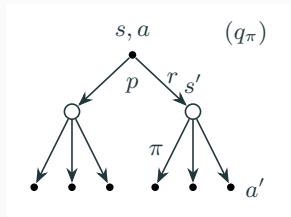
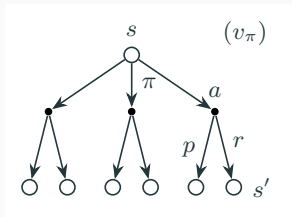


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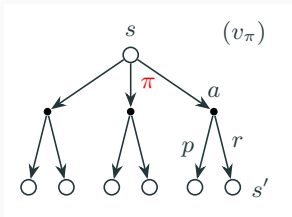


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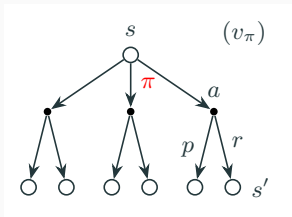


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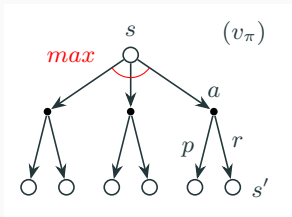


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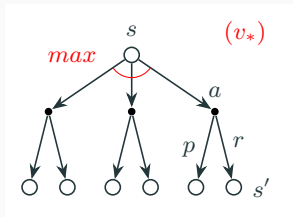


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Optimal Policies and Optimal Value Functions

Optimal Policy

π_* is said to be optimal if its expected return is greater than or equal to that of all other policies for all states:

$$v_{\pi_*}(s) \geq v_{\pi'}(s) \quad \forall s \in \mathcal{S}, \forall \text{ policies } \pi'$$

Optimal Value Functions

Value functions associated with optimal policies:

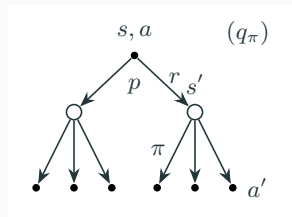
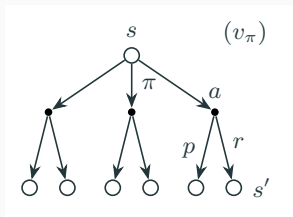
$$\begin{aligned} v_*(s) &= v_{\pi_*}(s) = \max_{\pi} v_{\pi}(s) & \forall s \in \mathcal{S} \\ q_*(s, a) &= q_{\pi_*}(s, a) = \max_{\pi} q_{\pi}(s, a) & \forall a \in \mathcal{A}(s), \forall s \in \mathcal{S} \end{aligned}$$

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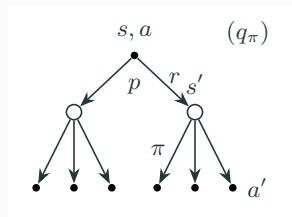
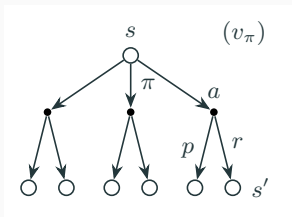


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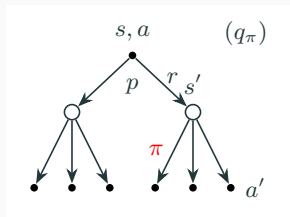
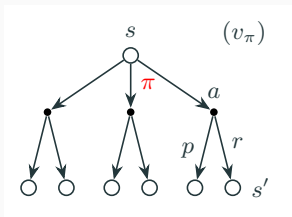


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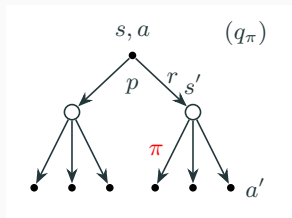
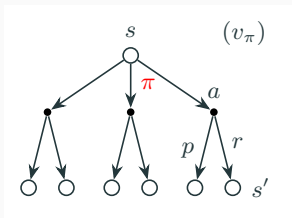


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Optimal Policies and Optimal Value Functions

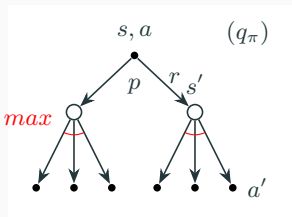
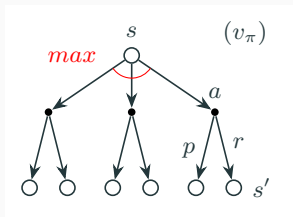
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Note: In the original image, red arrows point from $\pi(a | s)$ to $\max_a$ and from $\sum_{a' \in \mathcal{A}(s')}$ to $\max_{a'}$.

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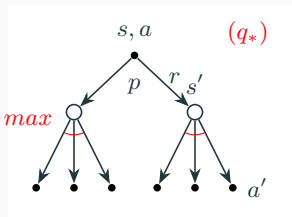
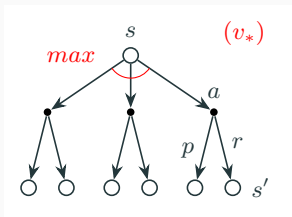
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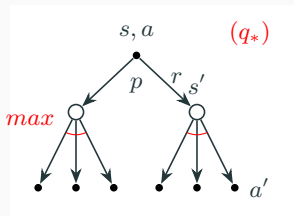
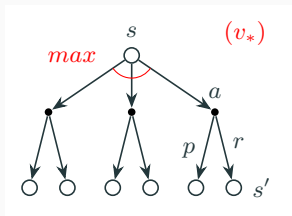


Optimal Policies and Optimal Value Functions

Bellman Optimality Equations

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$$q_*(s, a) = \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma \max_{a' \in \mathcal{A}(s')} q_*(s', a') \right]$$



Chapter 4

Dynamic Programming
